2024

01

Cardio Ring

Note: Some sketches and pictures of prototype were omitted due to confidentiality

ArteVu 2nd Generation

Location
Taiwan

Project Time
02/2024 –
Present

Project Brief
This is an early development project for the 2nd generation of the ArteVu, a cuffless, continuous, and non-invasive blood pressure monitor. The goal is to implement it seamlessly into the user's daily life.

My Role
I worked as user experience researcher/engineer in this project. I conducted user research and made iterations of prototypes for multiple rounds of user testing including the biggest one in Japan.

Overview

Continuous, Non-invasive Blood Pressure (CNBP)

Continuous blood pressure (CBP) is important because it provides a comprehensive picture of blood pressure fluctuation throughout the whole day. It can provide the clinician with important information to help detect patterns, evaluate treatment options and result, assess risk of cardiovascular diseases, and many other benefits. Overall CBP provides a more detailed and accurate understanding of the patient's blood pressure for better diagnosis and treatment.

Currently, the golden standard of CBP is measured through invasive arterial line catheter. However, it puts both the clinician and the patient at risk. Therefore, Cardio Ring is trying to create a novel device that measures CNBP through a proprietary mechanical sensor on the finger, for a simple and low-risk CNBP measurement solution.

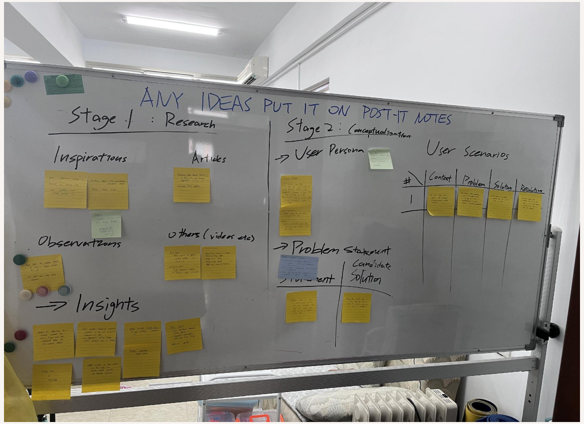
Design Process

Methods

- Online research
    - Read research papers, reports, and documents on CNBP, measurement methods, medical device design, and wearable design.
    - Articles and forums on what do the users like and dislike in wearable devices
- Field observation
- Stakeholder interview
    - Interview and questionnaires regarding ArteVu, other sphygmomanometers, and wearable devices, what do they like and how they would improve the product?
- Design jam
    - Visualize and organize perspectives and ideas
- Sketches
- User testing

Conclusion

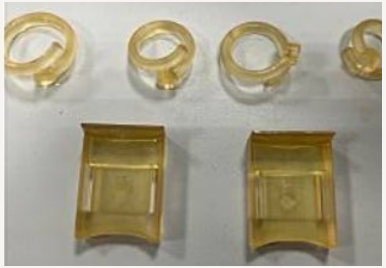
- A low-profile wearable device that can measure CNBP with minimal interference with the user's daily activities

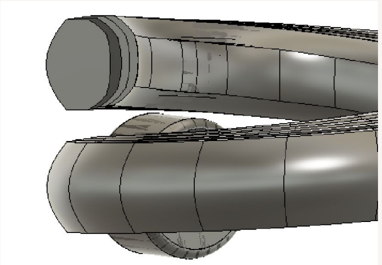


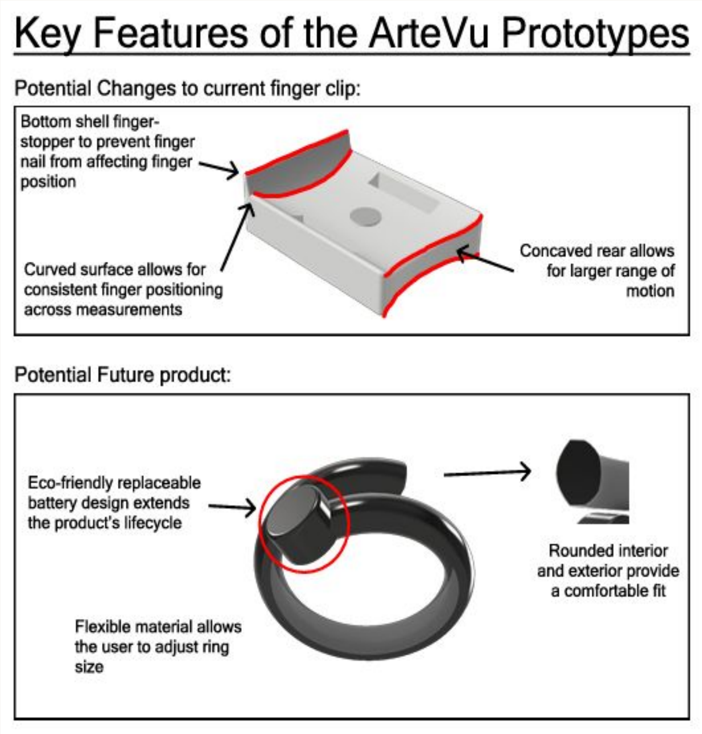
Prototypes

There were multiple iterations of the prototypes for the current generation of ArteVu and 2nd generation heavily focused on the improvement for fit, both for the comfort of the user and the consistency of sensor contact. The most difficult parts of this process were trying to find the most "universal" curvature, from the comfort groove on the finger clip to the ring band shape.

Below are pictures of early prototypes and the ring bands shape stacked together.

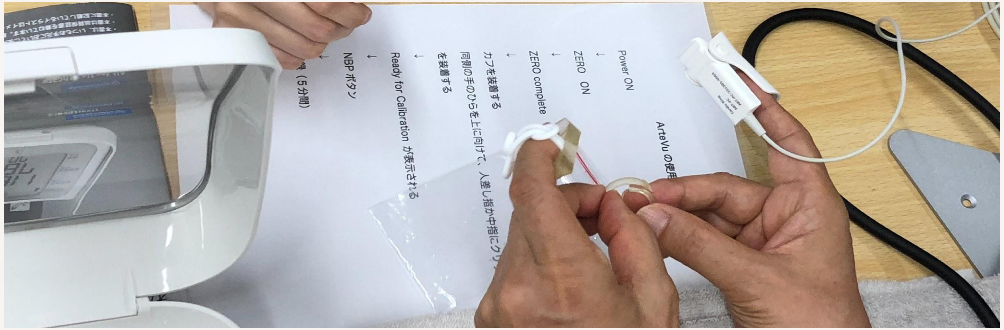






User Testing

These photos are from the user testing we did at Kyorin University in Japan, on the current ArteVu and new prototypes.



Future and Learning Outcomes

Future Goals

- Through multiple rounds of user testing, the results yield the ring prototype as favorite
- Study the implementation of the mechanical sensor into a ring, along with the rest of the components such as Bluetooth module
- Look into more features the product can have with the biomarkers currently measured
    - Stress score calculated from heartrate variability

Learning Outcomes

- Experience conducting user research in a professional setting
- Experience designing and prototyping product with focus in usability and sustainability
- Knowledge in regulations and standards regarding medical devices and sphygmomanometer in particular